Incident Report: (NatureForce Exercise)

1. Summary

On October 1, 2026 at 19:30:36 UTC, the Windows host 10.10.1[.]128 (DESKTOP-4VGYQX7, user bdiaz) downloaded a 1,646,592-byte executable named runner_ilove.exe from the external server 107.175.82[.]242 over unencrypted HTTP on TCP port 9000, using the command-line tool curl. The download completed, so the SOC alert for an EXE sent over an unusual port is a **true positive**.

The second alert, for command-and-control (C2) traffic to 195.64.128[.]106:443, **could not be confirmed** because the capture contains no traffic to or from that address.

Severity: treat as high priority until endpoint logs show the file did not execute.

Action: isolate the host, identify what launched curl, and block 107.175.82[.]242.

2. Scope and method

Source: public training exercise "Traffic Analysis Exercise: NatureForce" dated 2026-10-01, from Malware-Traffic-Analysis.net. All data is from a simulated environment. The scenario: a SOC analyst sees two alerts and reviews a packet capture from the associated Windows host.

Alerts under review

- 107.175.82[.]242:9000: Windows executable (EXE) sent from an IP address over an unusual TCP port
- 195.64.128[.]106:443: CNCmachineRMS RAT C2 traffic

Environment

Item	Value
LAN segment	10.10.1[.]0/24
Domain	natureforce[.]com
Domain controller	10.10.1[.]10 (WIN-LEKBU2OY51N)
Gateway	10.10.1[.]1

Method: Using Wireshark to capture packets, I identified the infected host, followed the HTTP download, extracted the file for hashing, checked both alerts against the traffic, reviewed activity in the seconds before the download, and mapped observed behavior to MITRE ATT&CK. Times are in UTC and the capture start (about 19:27:27 UTC) is derived from packet timestamps.

3. Affected asset

Attribute	Value	How identified
IP address	10.10.1[.]128	Source of the HTTP request in packet 23091
MAC address	00:22:fb:ec:7e:e7	Ethernet source field of the host's packets (vendor prefix: Intel)
Host name	DESKTOP-4VGYQX7	Taken from the host's network traffic in the capture
User account	bdiaz	Taken from the host's network traffic in the capture

4. Timeline (UTC, October 1, 2026)

Times after the first row are derived from the capture's packet timestamps and are accurate to about a tenth of a second.

Time (UTC)	Event	Evidence
18:46:36 (server-reported)	Server reports the file was last modified. This comes from a response header and was not observed in the capture.	Last-Modified header, packet 24916
19:30:29.83	DC 10.10.1[.]10 answers the host's DNS query for vsblobprodscussu5shard51[.]blob[.]core[.]windows[.]net with 57.150.53[.]97 and two other addresses	Packets 10187 and 10188
19:30:29.86	Host opens a TCP connection to 57.150.53[.]97:443	Packet 10190
19:30:29.91	TLS Client Hello with SNI vsblobprodscussu5shard51[.]blob[.]core[.]windows[.]net	Packet 10195
19:30:36.48	Host begins closing that TLS session after about 11.5 MB was received	Packet 23079

Time (UTC)	Event	Evidence
19:30:36.73	Host opens a TCP connection to 107.175.82[.]242:9000	Packets 23088 to 23090
19:30:36.81	Host sends GET /wilow/runner_ilove.exe using curl/8.21.0	Packet 23091
19:30:38.58	Last packet of the server's 200 OK response; all 1,646,592 bytes received (1,216 TCP segments)	Packet 24916
After 19:30:38	No further traffic between the host and 107.175.82[.]242 in the capture	Analyst review

5. Findings

Finding 1: Executable download confirmed (alert 1 is a true positive).

Packet 23091 is an HTTP request from 10.10.1[.]128 to 107.175.82[.]242:9000 for `/wilow/runner_ilove.exe`. The request has `Host: 107.175.82[.]242:9000`, `User-Agent: curl/8.21.0`. The server answered with `200 OK`, content type `binary/octet-stream` and a `Content-Length` of 1,646,592 bytes in packet 24916. Wireshark reassembled exactly 1,646,592 bytes of file data from 1,216 TCP segments, and the TCP stream is marked complete with data. The transfer took about 1.8 seconds.

The extracted file has the SHA-256 `b789adce70c028a6d17be45ec34667aa95586350ee0c4057e75b138b1ad35061`.

The curl User-Agent is consistent with a tool-based download rather than a person clicking a link in a browser. A User-Agent can be faked, and the capture does not show what started curl.

Finding 2: C2 alert not confirmed (alert 2).

A display filter of `ip.addr == 195.64.128.106` returned zero packets. The capture contains no traffic to or from 195.64.128[.]106, so the CNCmachineRMS RAT C2 alert could not be verified. This does not rule out C2 activity after the capture ended.

6. Indicators of compromise

All values are defanged.

Type	Value	Context
IP and port	107.175.82[.]242:9000	Server that delivered the EXE

Type	Value	Context
URL	hxxp://107.175.82[.]242:9000/wilow/runner_ilove.exe	Download URL
File name	runner_ilove.exe	Downloaded executable
SHA-256	b789adce70c028a6d17be45ec34667aa95586350ee0c4057e75b138b1ad35061	Hash of the extracted file
File size	1,646,592 bytes	Content-Length, matches reassembled data

7. MITRE ATT&CK mapping

Only behavior with evidence in the capture is mapped.

Tactic	Technique	Evidence
Command and Control	Ingress Tool Transfer (T1105)	The host downloaded an executable from an external server (packets 23091 to 24916)
Command and Control	Non-Standard Port (T1571)	HTTP traffic carried over TCP port 9000 instead of port 80

8. Conclusion

The host 10.10.1[.]128 (DESKTOP-4VGYQX7, user bdiaz) downloaded a Windows executable from 107.175.82[.]242:9000 at 19:30:36 UTC on October 1, 2026. The download completed, so the EXE alert is a true positive.

Question	Assessment	Confidence
Was the EXE delivered to the host?	Yes, all 1,646,592 bytes	High
Was it fetched by a tool rather than a browser?	Consistent with curl	Medium (User-Agent can be faked)
Is there C2 traffic to 195.64.128[.]106?	Not seen in this capture	High that it is absent from the capture; unknown beyond it
Did the file execute?	Cannot be determined	Not assessed

Because delivery is confirmed but execution is not, the host should be treated as potentially compromised until endpoint evidence shows otherwise.

9. Recommended response

Contain (now)

1. Isolate host 10.10.1[.]128 from the network.
2. Block 107.175.82[.]242 at the firewall.
3. Also block 195.64.128[.]106 as a precaution because it appears in the C2 alert.

Investigate

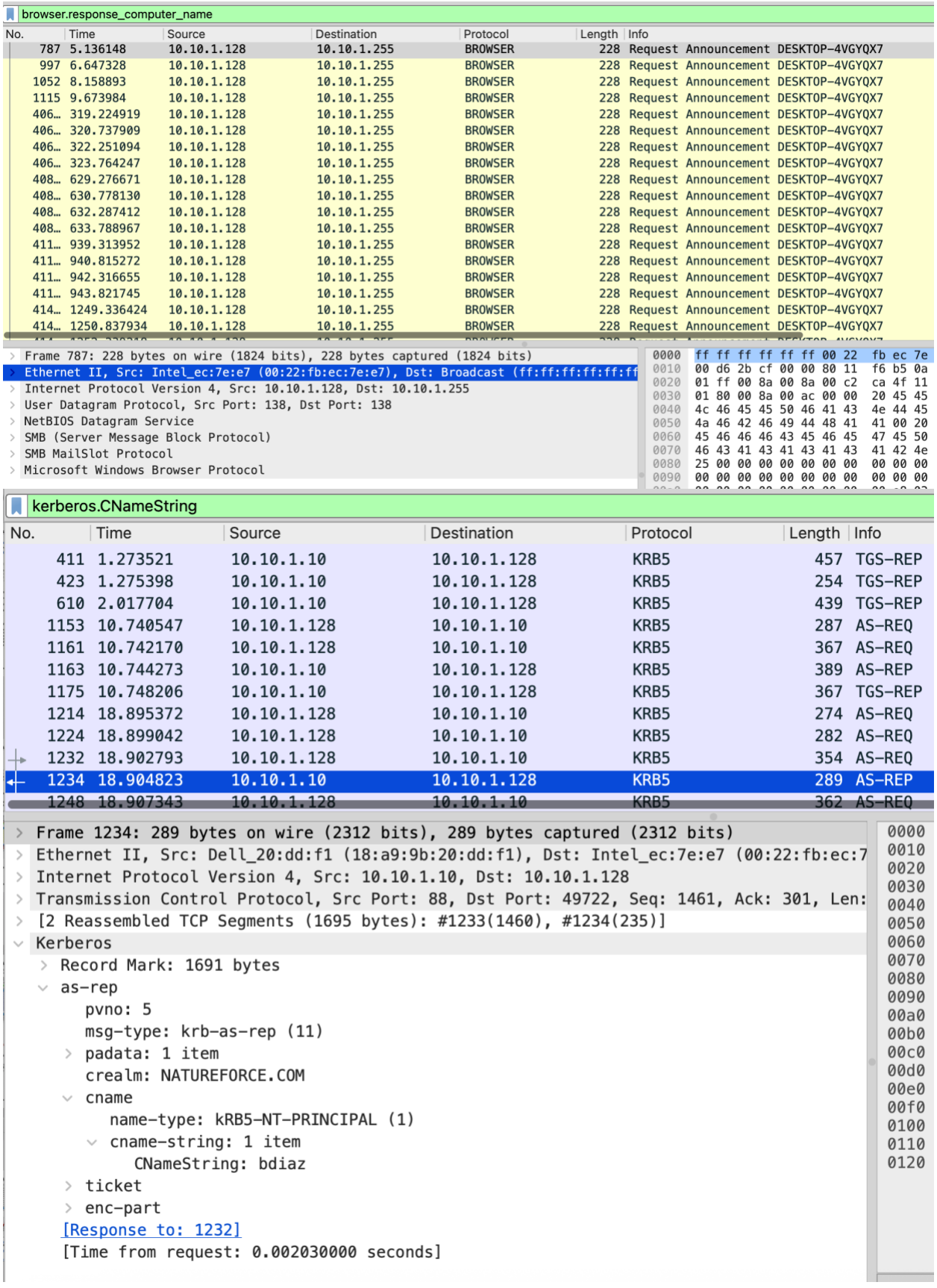
4. Pull endpoint or EDR logs for the host around 19:30 UTC to find which process started curl.
5. Check whether runner_ilove.exe exists on disk and whether it ran and look for persistence such as new services or scheduled tasks.
6. Search firewall, proxy and DNS logs for other hosts that contacted 107.175.82[.]242 or 195.64.128[.]106.
7. Review authentication logs for the account bdiaz and the domain controller for unusual activity.
8. Analyze the file safely in a sandbox using the SHA-256, never on a production machine.

Eradicate and recover

9. Reset bdiaz's credentials and any other credentials used on the host.
10. If execution or persistence is confirmed, re-image the host before returning it to service.

Prevent

11. Alert on executable downloads over non-standard ports and on curl fetching executables.
12. Restrict outbound traffic to approved ports and destinations where possible.
13. Review how the user or attacker could run curl, for example application control and script restrictions.


browser.response_computer_name
No. Time Source Destination Protocol Length Info
787 5.136148 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
997 6.647328 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
1052 8.158893 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
1115 9.673984 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
406… 319.224919 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
406… 320.737909 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
406… 322.251094 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
406… 323.764247 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
408… 629.276671 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
408… 630.778130 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
408… 632.287412 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
408… 633.788967 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
411… 939.313952 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
411… 940.815272 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
411… 942.316655 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
411… 943.821745 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
414… 1249.336424 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
414… 1250.837934 10.10.1.128 10.10.1.255 BROWSER 228 Request Announcement DESKTOP-4VGYQX7
Frame 787: 228 bytes on wire (1824 bits), 228 bytes captured (1824 bits)
Ethernet II, Src: Intel_ec:7e:e7 (00:22:fb:ec:7e:e7), Dst: Broadcast (ff:ff:ff:ff:ff:ff
Internet Protocol Version 4, Src: 10.10.1.128, Dst: 10.10.1.255
User Datagram Protocol, Src Port: 138, Dst Port: 138
NetBIOS Datagram Service
SMB (Server Message Block Protocol)
SMB MailSlot Protocol
Microsoft Windows Browser Protocol
0000 ff ff ff ff ff ff 00 22 fb ec 7e
0010 00 d6 2b cf 00 00 80 11 f6 b5 0a
0020 01 ff 00 8a 00 8a 00 c2 ca 4f 11
0030 01 80 00 8a 00 ac 00 00 20 45 45
0040 4c 46 45 45 50 46 41 43 4e 44 45
0050 4a 46 42 46 49 44 48 41 41 00 20
0060 45 46 46 46 43 45 46 45 47 45 50
0070 46 43 41 43 41 43 41 43 41 42 4e
0080 25 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00
kerberos.CNameString
No. Time Source Destination Protocol Length Info
411 1.273521 10.10.1.10 10.10.1.128 KRB5 457 TGS-REP
423 1.275398 10.10.1.10 10.10.1.128 KRB5 254 TGS-REP
610 2.017704 10.10.1.10 10.10.1.128 KRB5 439 TGS-REP
1153 10.740547 10.10.1.128 10.10.1.10 KRB5 287 AS-REQ
1161 10.742170 10.10.1.128 10.10.1.10 KRB5 367 AS-REQ
1163 10.744273 10.10.1.10 10.10.1.128 KRB5 389 AS-REP
1175 10.748206 10.10.1.10 10.10.1.128 KRB5 367 TGS-REP
1214 18.895372 10.10.1.128 10.10.1.10 KRB5 274 AS-REQ
1224 18.899042 10.10.1.128 10.10.1.10 KRB5 282 AS-REQ
1232 18.902793 10.10.1.128 10.10.1.10 KRB5 354 AS-REQ
1234 18.904823 10.10.1.10 10.10.1.128 KRB5 289 AS-REP
1248 18.907343 10.10.1.128 10.10.1.10 KRB5 362 AS-REQ
Frame 1234: 289 bytes on wire (2312 bits), 289 bytes captured (2312 bits)
Ethernet II, Src: Dell_20:dd:f1 (18:a9:9b:20:dd:f1), Dst: Intel_ec:7e:e7 (00:22:fb:ec:7
Internet Protocol Version 4, Src: 10.10.1.10, Dst: 10.10.1.128
Transmission Control Protocol, Src Port: 88, Dst Port: 49722, Seq: 1461, Ack: 301, Len:
[2 Reassembled TCP Segments (1695 bytes): #1233(1460), #1234(235)]
Kerberos
Record Mark: 1691 bytes
as-rep
pvno: 5
msg-type: krb-as-rep (11)
padata: 1 item
crealm: NATUREFORCE.COM
cname
name-type: kRB5-NT-PRINCIPAL (1)
cname-string: 1 item
CNameString: bdiaz
ticket
enc-part
[Response to: 1232]
[Time from request: 0.002030000 seconds]
0000
0010
0020
0030
0040
0050
0060
0070
0080
0090
00a0
00b0
00c0
00d0
00e0
00f0
0100
0110
0120